

ECSE-4040

Digital Electronics

Lecture 19

Bipolar Digital Circuits

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References: Appendix B: HJS

Chapters 5 and 17, Neamen

<https://lms.rpi.edu>

Lecture Outline

- ECL Basic Concepts
- DC Analysis
- Fanout
- Modified Low Power ECL
- Series Gating

High Speed Logic Families

Emitter-Coupled Logic (ECL)

- Bipolar
- III-V FET
- BiCMOS



Current-Mode Logic (CML)

- ☐ Bipolar
- ☐ MOS
- ☐ BiCMOS



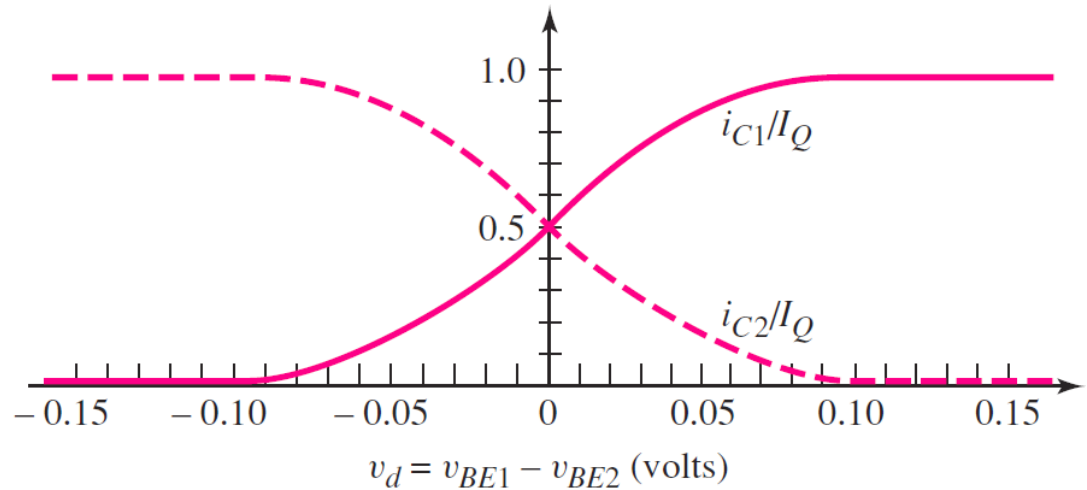
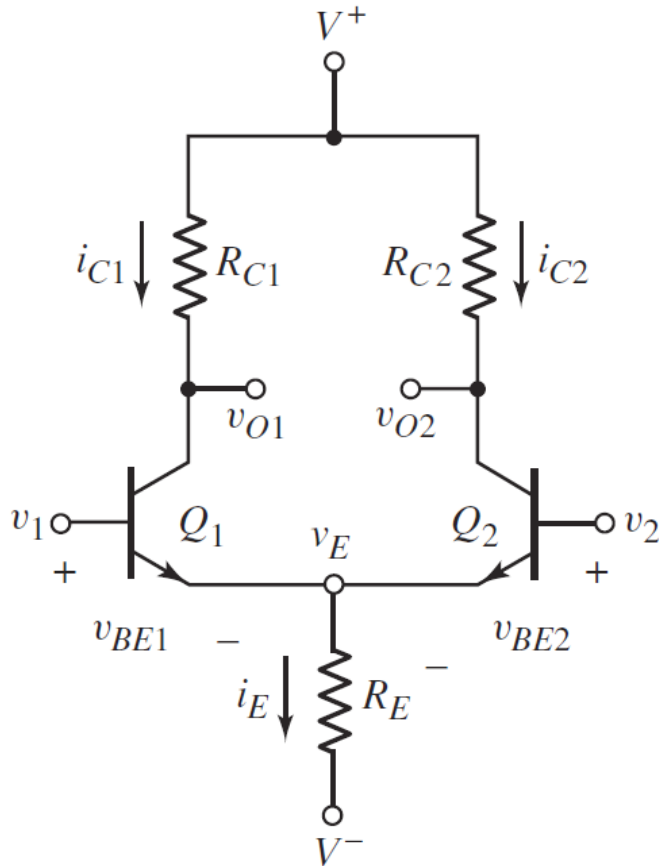
**Some high-speed logic families use differential topologies

Emitter Coupled Logic

- The emitter-coupled logic (ECL) circuit is based on the differential amplifier circuit,
- In digital applications, the diff-amp is driven into its nonlinear region.
- The transistors are either cut off or in the active region.
- Saturation is avoided in order to minimize switching times and propagation delay times.
- ECL circuits have the shortest propagation delay times of any bipolar digital technology.

ECL-Differential Amplifiers

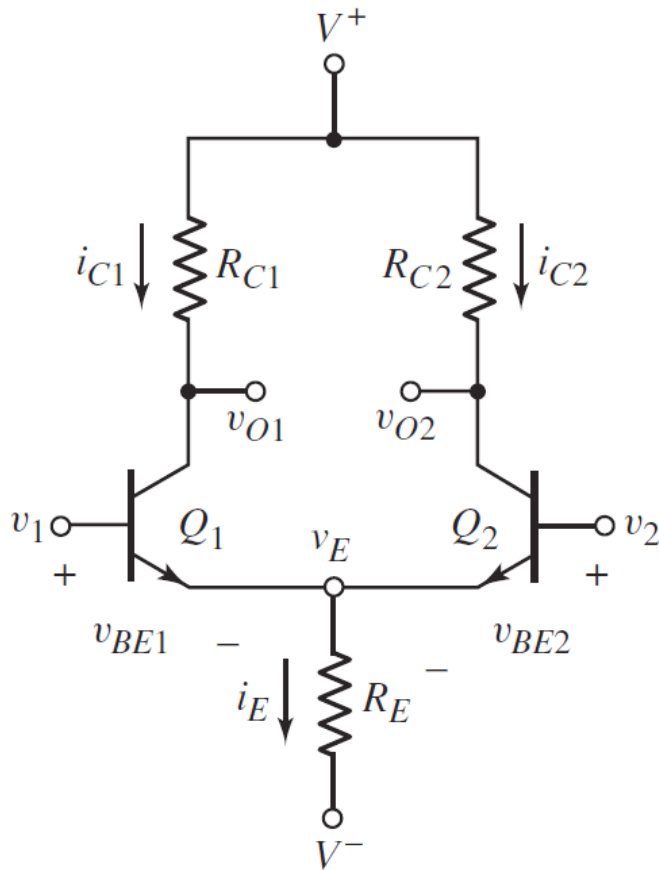
Revisited



$$i_{C1} = I_S e^{v_{BE1}/V_T}$$

$$i_{C2} = I_S e^{v_{BE2}/V_T}$$

ECL-Basic Concepts

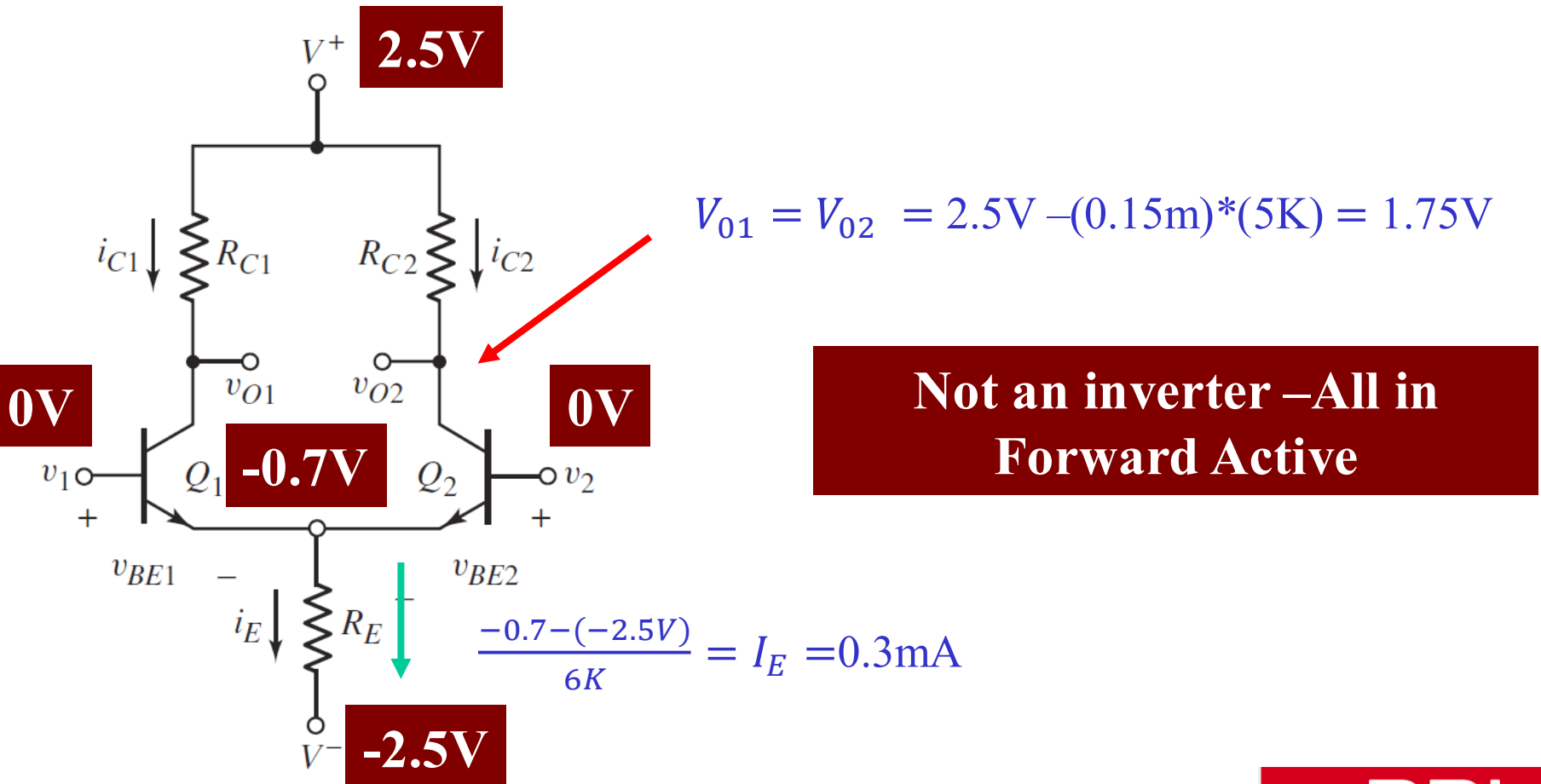


$$\frac{I_{C1}}{I_{C2}} = \frac{e^{\left(\frac{V_{BE1}}{V_T}\right)}}{e^{\left(\frac{V_{BE2}}{V_T}\right)}} = e^{\left(\frac{V_{BE1} - V_{BE2}}{V_T}\right)}$$

ΔV_{BE}	$\frac{I_{C1}}{I_{C2}}$
0.078	20.08554
0.104	54.59815
0.12	101.0267
0.13	148.4132

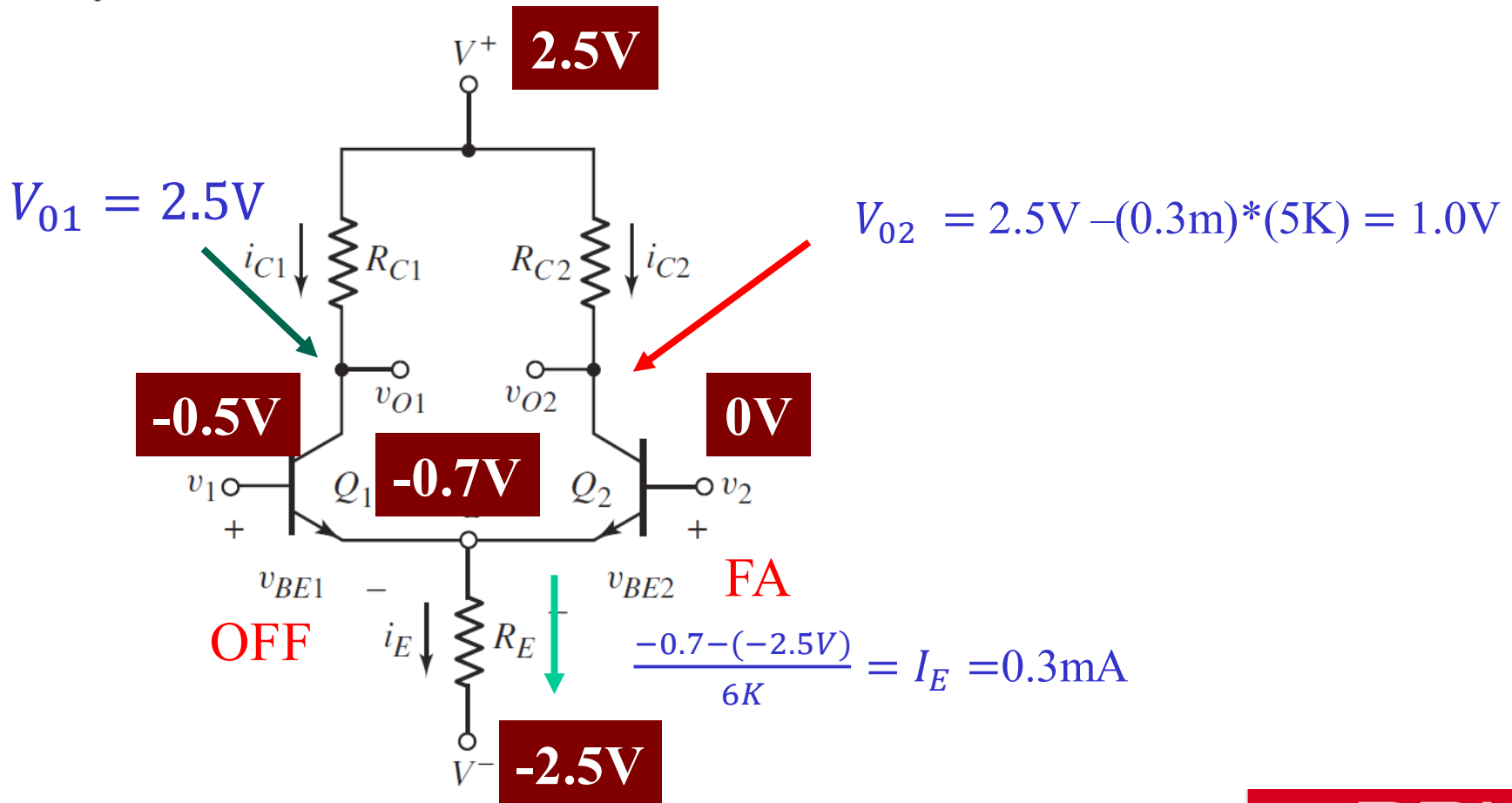
ECL-Example DC Analysis

$R_{C1} = R_{C2} \equiv R_C = 5\text{ k}\Omega$, $R_E = 6\text{ k}\Omega$, and $v_2 = 0$. Neglect base currents in the dc analysis.



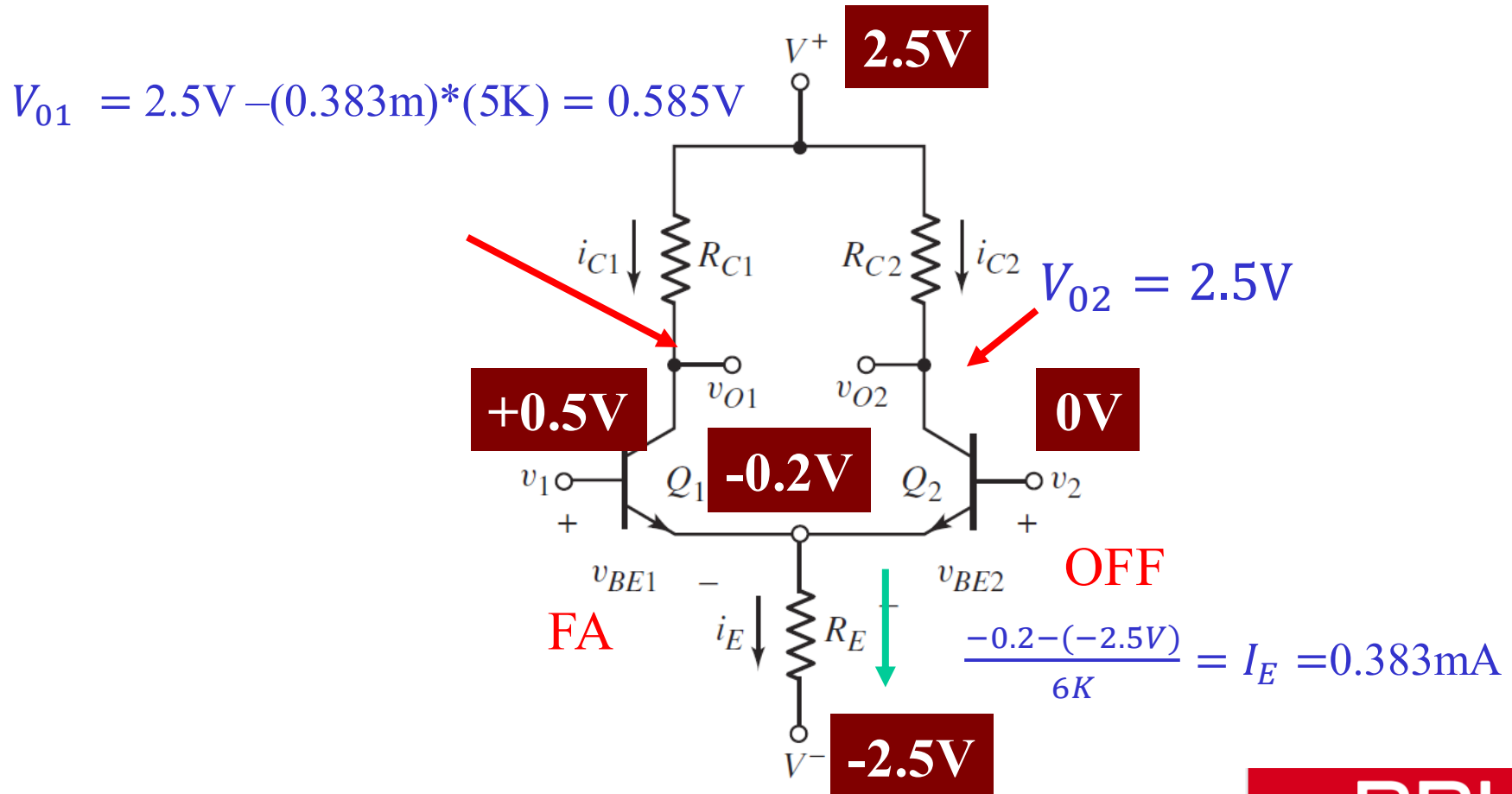
ECL-Example DC Analysis

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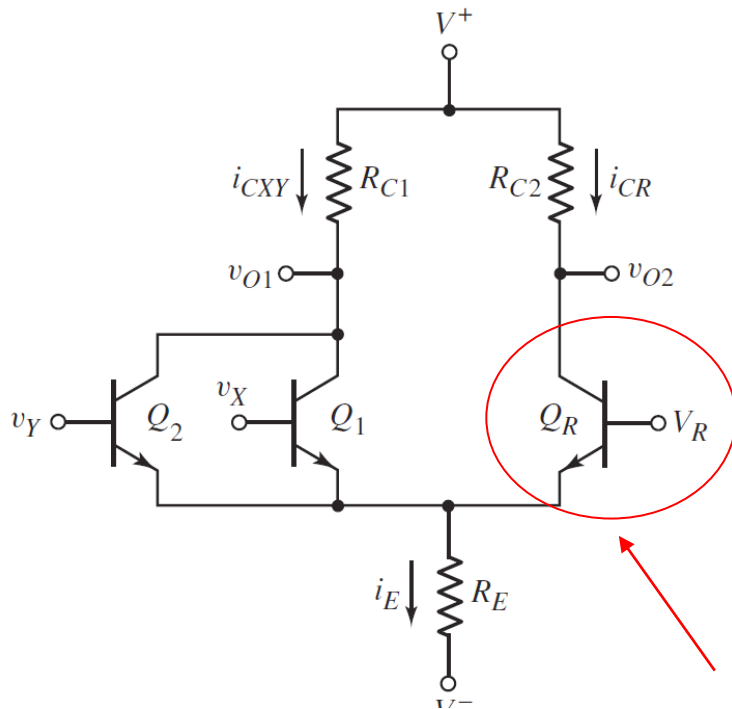
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$R_{C1} = R_{C2} \equiv R_C = 5\text{ k}\Omega$, $R_E = 6\text{ k}\Omega$, and $v_2 = 0$. Neglect base currents in the dc analysis.



ECL Basic Logic

NOR
↓
OR
↓

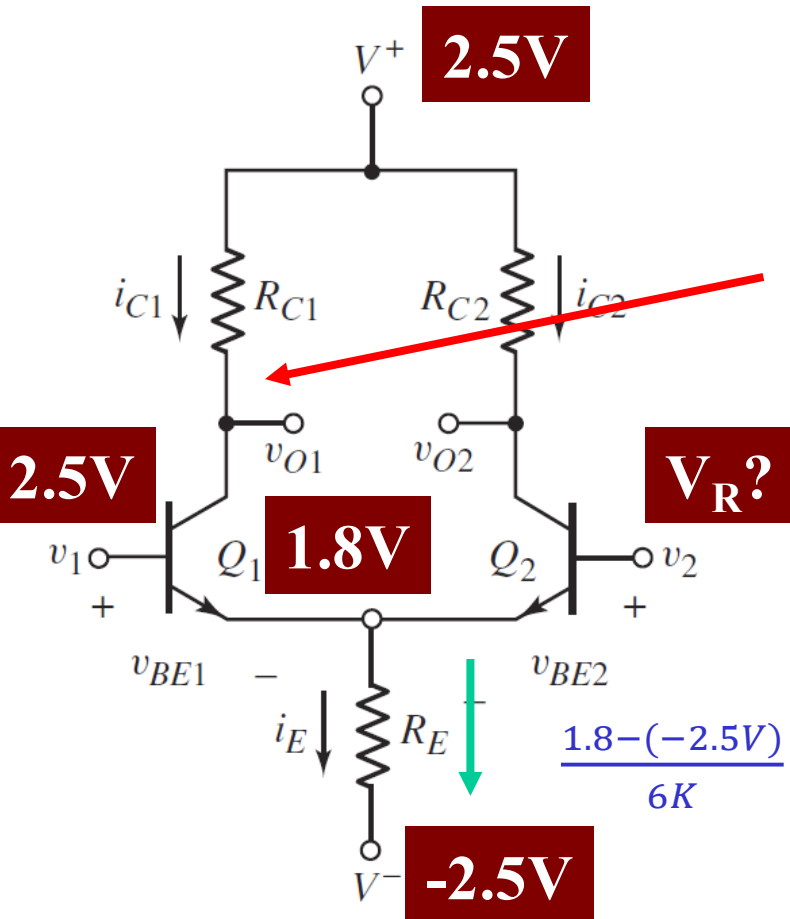


Reference transistor

		Q1	Q2	Q _R	V _{O1}	V _{O2}
$V_x < V_R$	$V_y < V_R$	OFF	OFF	ON	V+	?
$V_x > V_R$	$V_y < V_R$	ON	OFF	OFF	?	V+
$V_x < V_R$	$V_y > V_R$	OFF	ON	OFF	?	V+
$V_x > V_R$	$V_y > V_R$	ON	ON	OFF	?	V+

What happens when you drive another gate?

$R_{C1} = R_{C2} \equiv R_C = 5\text{ k}\Omega$, $R_E = 6\text{ k}\Omega$, and $v_2 = 0$. Neglect base currents in the dc analysis.

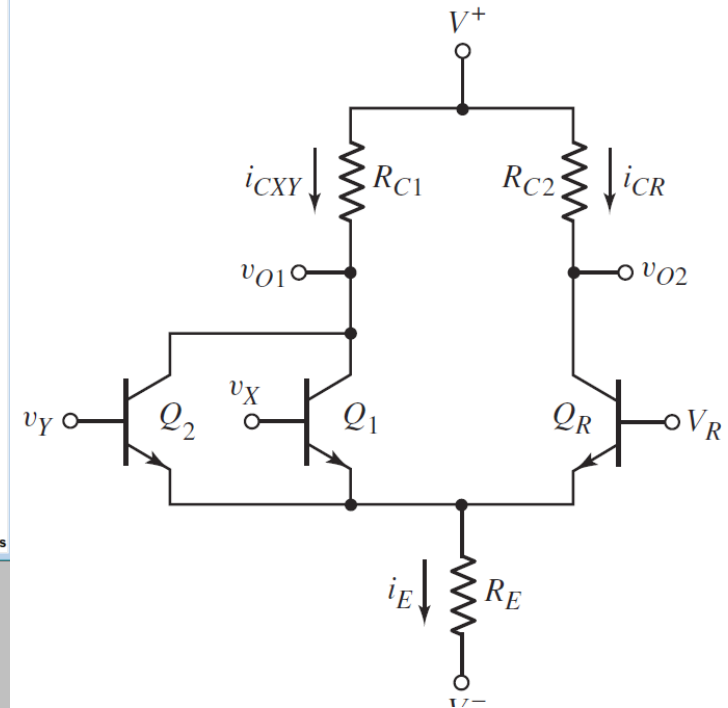
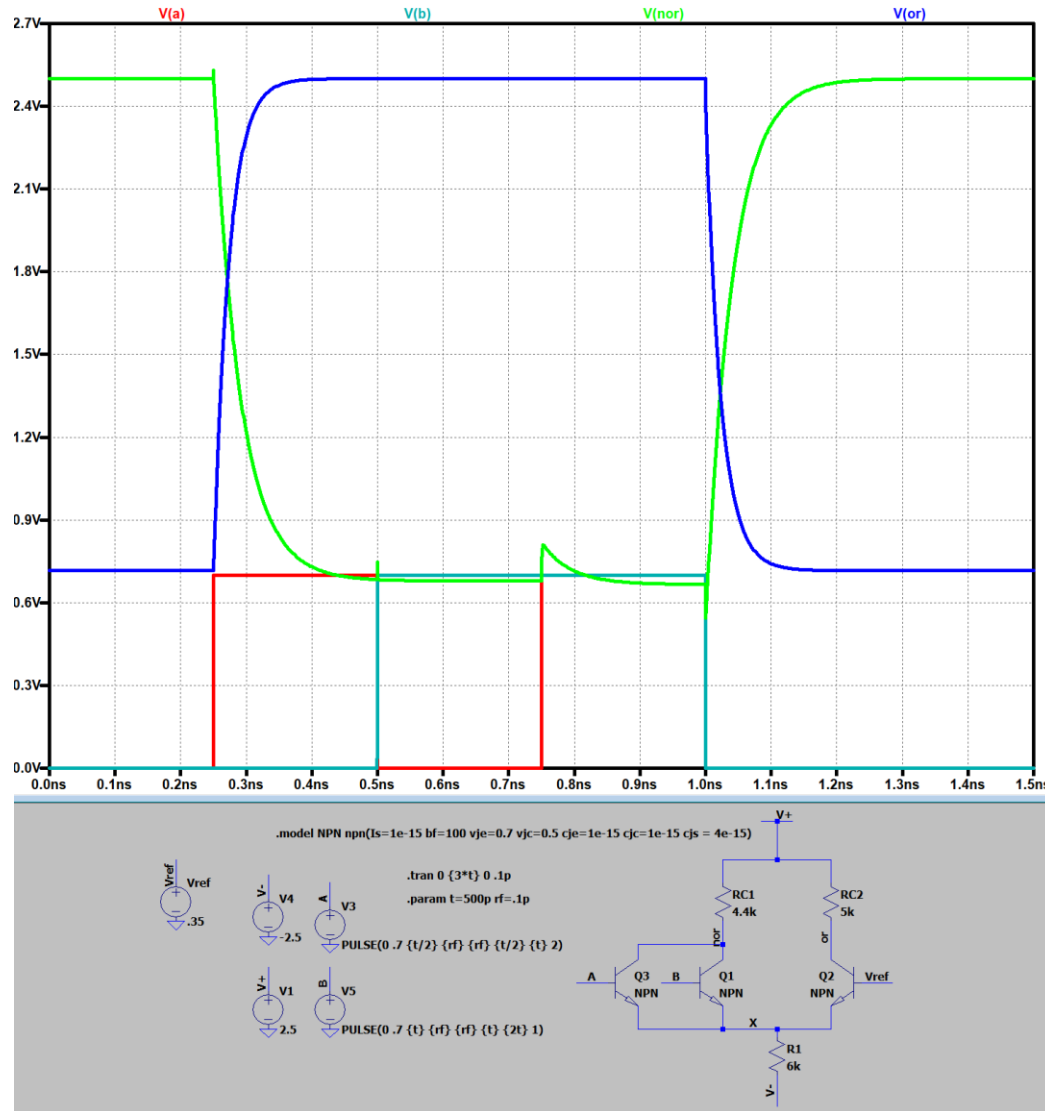


$$V_{O1} = 2.5\text{V} - (0.716\text{m}) \cdot (5\text{K}) = -1.083\text{V}$$

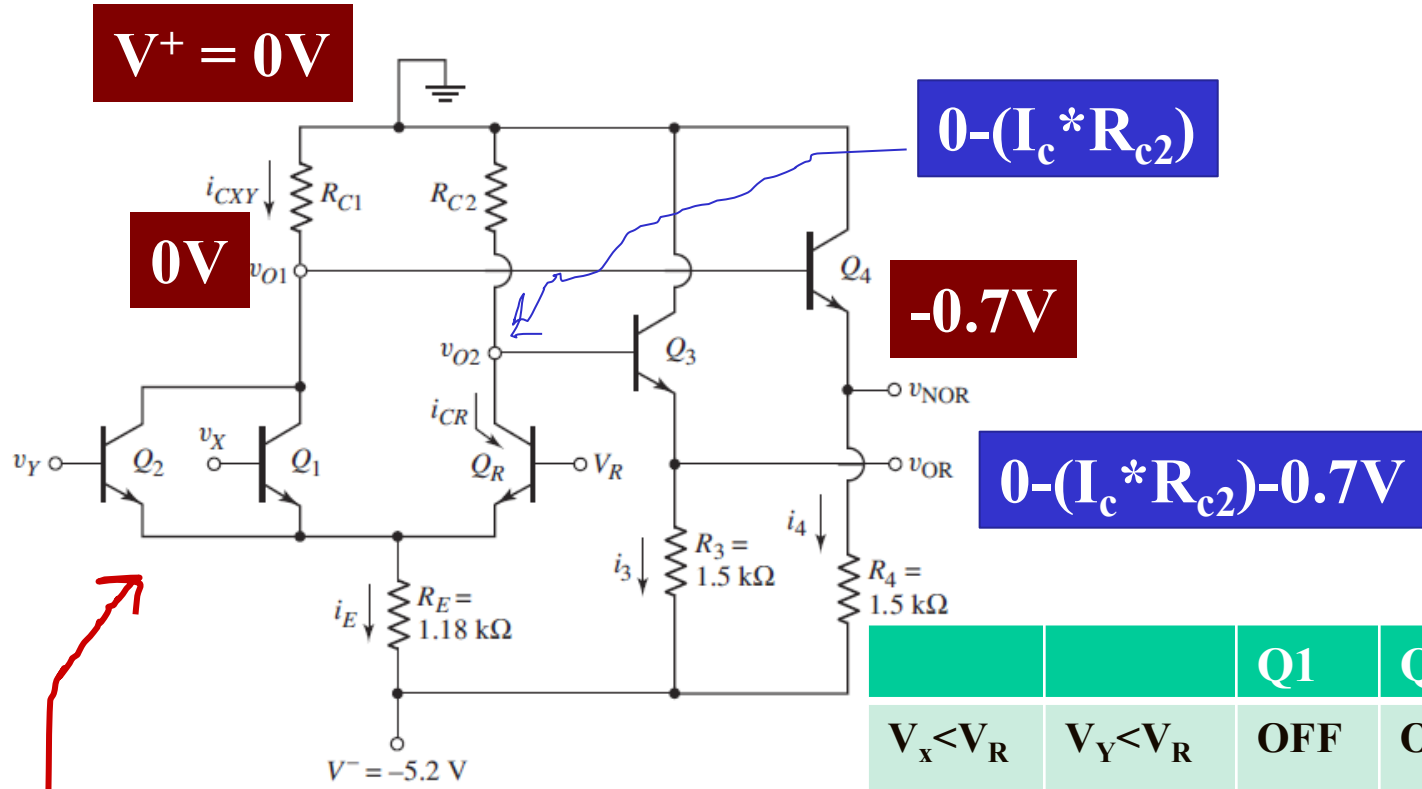
What will happen to the mode of the ON device?

$$\frac{1.8 - (-2.5\text{V})}{6\text{K}} = I_E = 0.716\text{mA}$$

ECL Basic Logic



ECL With Emitter Followers (logic 1)

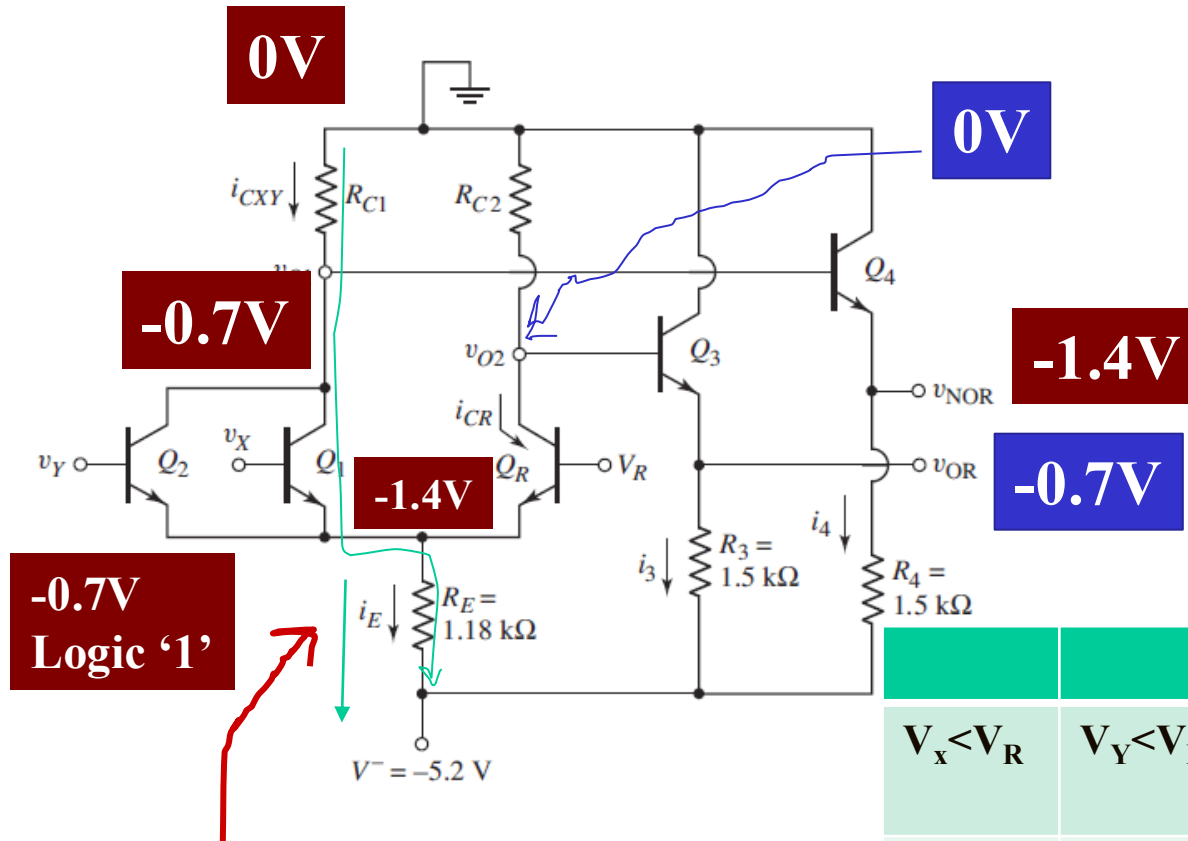


NOR **OR**

↓ ↓

		Q1	Q2	QR	V _{O1}	V _{O2}
$V_x < V_R$	$V_y < V_R$	OFF	OFF	ON	$V^+ - 0.7$	
$V_x > V_R$	$V_y < V_R$	ON	OFF	OFF		V^+
$V_x < V_R$	$V_y > V_R$	OFF	ON	OFF		V^+
$V_x > V_R$	$V_y > V_R$	ON	ON	OFF		V^+

ECL With Emitter Followers (logic 0)



$$\frac{-1.4 - (-5.2)}{1.18\text{K}} = 3.22\text{mA}$$

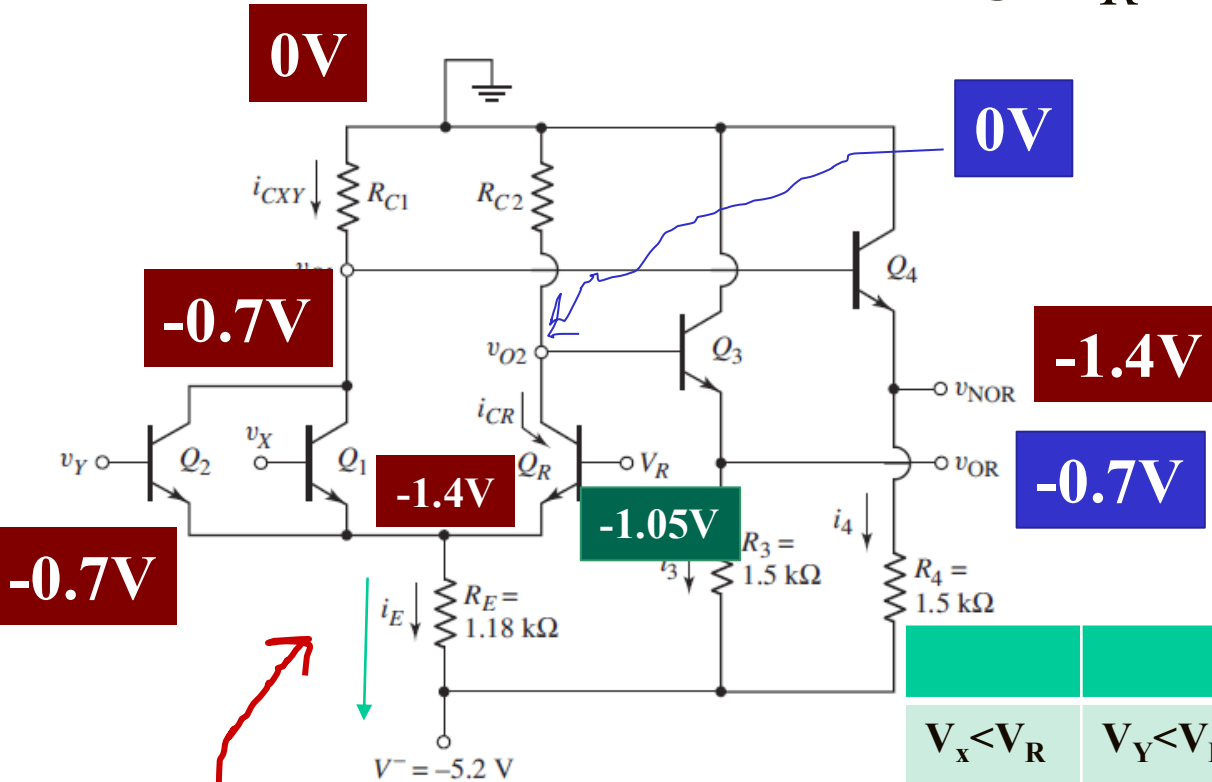
Use $RC1$ to set $V_{BC}=0$ to avoid transistor going to saturation (B-C junction at the onset of forward bias)

NOR OR

↓ ↓

		Q1	Q2	QR	Vo1	V02
$V_X < V_R$	$V_Y < V_R$	OFF	OFF	ON	V+ -0.7	-1.4
$V_X > V_R$	$V_Y < V_R$	ON	OFF	OFF	-1.4	V+ -0.7
$V_X < V_R$	$V_Y > V_R$	OFF	ON	OFF	-1.4	V+ -0.7
$V_X > V_R$	$V_Y > V_R$	ON	ON	OFF	-1.4	V+ -0.7

Finding V_R ?

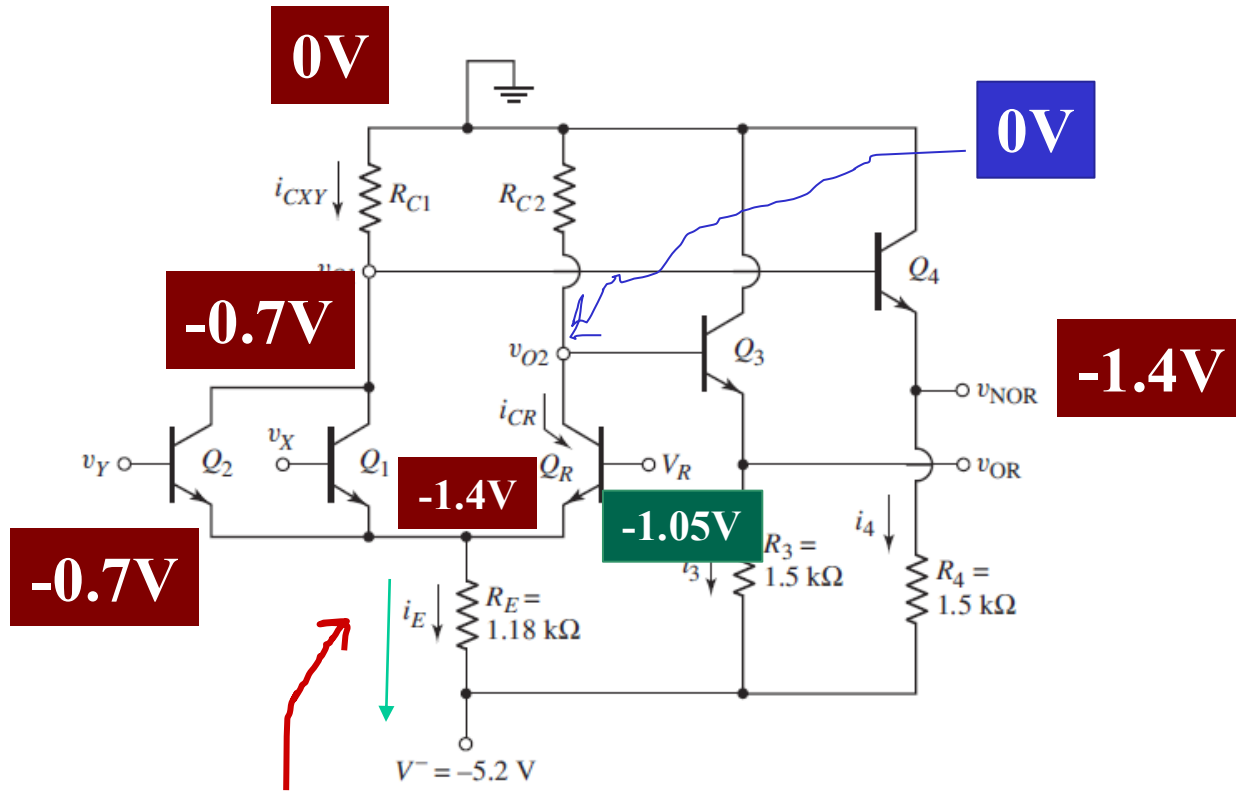


$$\frac{-1.4 - (-5.2)}{1.18K} = 3.22mA$$

Set V_R in between high and low values
 350mV in between high and low ($>$
 $120\text{nm } V$, which is $5V_T$)

		Q1	Q2	QR	Vo1	V02
$V_x < V_R$	$V_y < V_R$	OFF	OFF	ON	V+ -0.7	-1.4
$V_x > V_R$	$V_y < V_R$	ON	OFF	OFF	-1.4	V+ -0.7
$V_x < V_R$	$V_y > V_R$	OFF	ON	OFF	-1.4	V+ -0.7
$V_x > V_R$	$V_y > V_R$	ON	ON	OFF	-1.4	V+ -0.7

ECL With Emitter Followers: R_{c1} ?

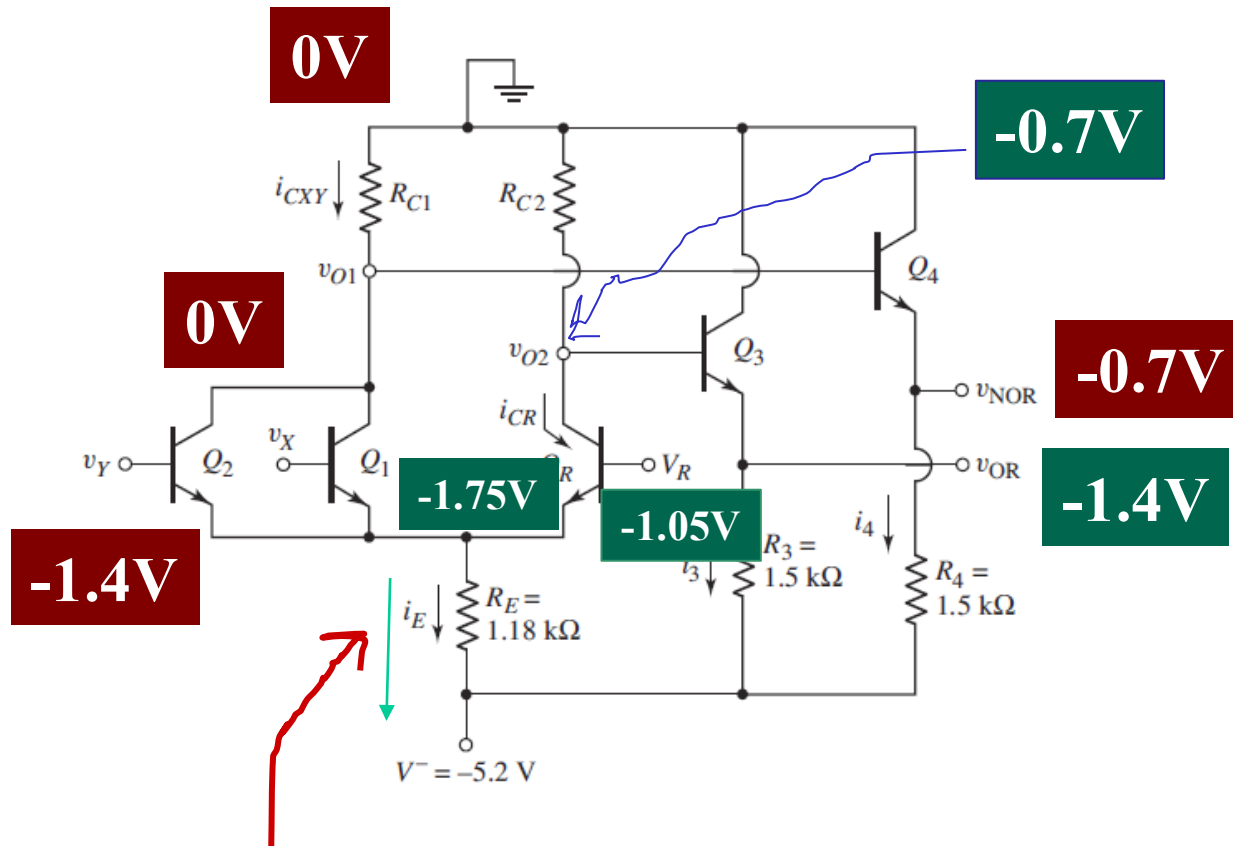


$$\frac{-1.4 - (-5.2)}{1.18K} = 3.22mA$$

$$R_{c1} = \frac{0 - (-0.7)}{3.22m} = 0.217K\Omega$$

Assume $\beta > 100$, and neglect base current

ECL With Emitter Followers: R_{c2} ?



$$\frac{-1.75 - (-5.2)}{1.18K} = 2.92\text{mA}$$

$$R_{c2} = \frac{0 - (-0.7)}{2.92\text{m}} = 0.2397K\Omega$$

Assume $\beta > 100$, and neglect base current

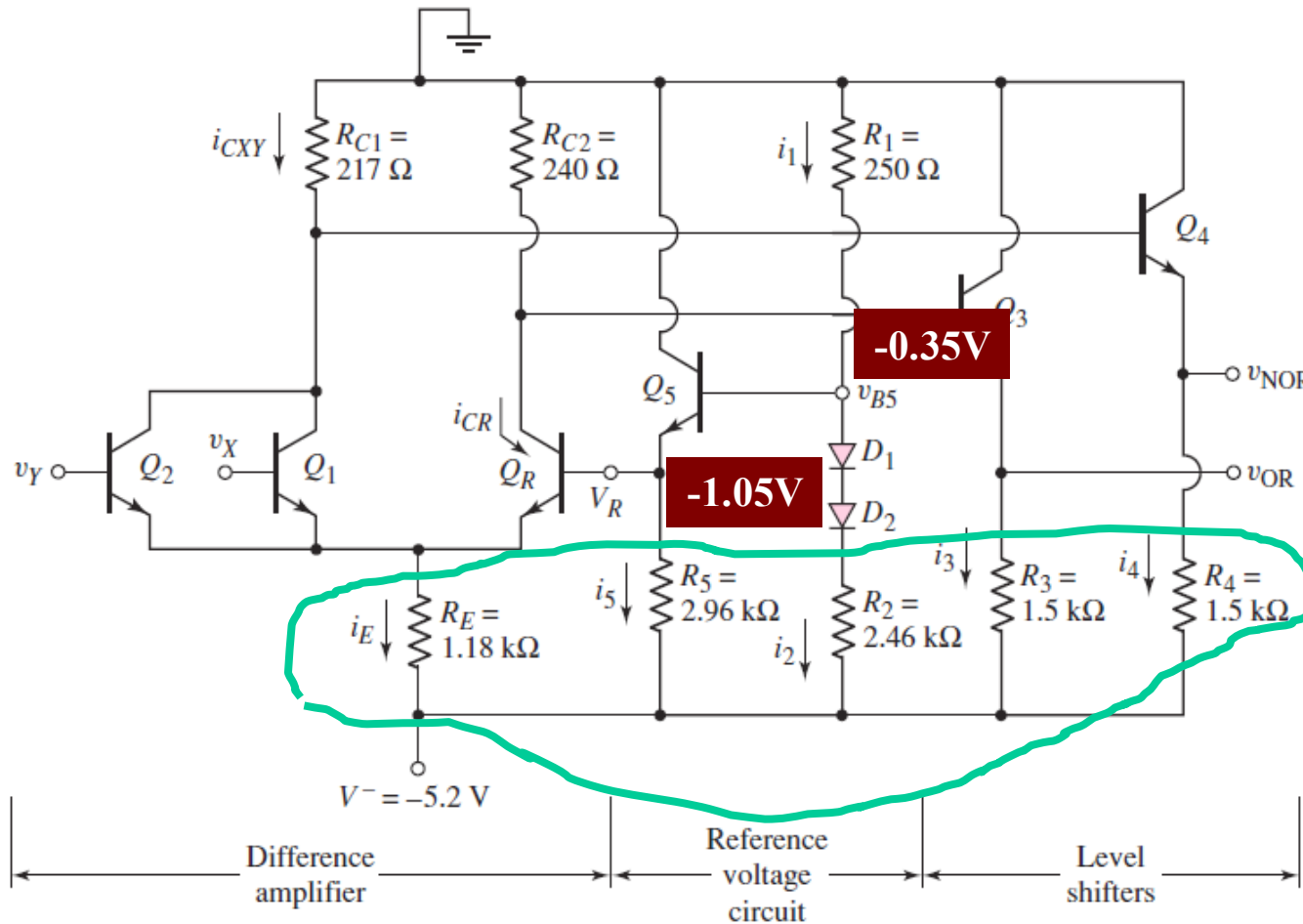
$R_{c1} \neq R_{c2}$ For symmetrical outputs

The diagram shows the timing relationship between several signals over a 30ps period. The signals are:

- V(or_pre)** (Red): Starts at -0.7V, transitions to -0.1V at 5ps, and back to -0.7V at 20ps.
- V(nor_pre)** (Cyan): Starts at 0.0V, transitions to -0.6V at 5ps, and back to 0.0V at 20ps.
- V(nor)** (Green): Starts at -0.7V, transitions to -1.4V at 5ps, and back to -0.7V at 20ps.
- V(or)** (Blue): Starts at -1.4V, transitions to -0.7V at 5ps, and back to -1.4V at 20ps.
- V(a)** (Magenta): A square wave between -0.7V and -1.4V, with transitions at 5ps, 10ps, 15ps, and 20ps.
- V(b)** (Black): A staircase signal that steps up at 10ps and 15ps, and steps down at 20ps.



Generating the Reference voltage



Design problem:
start by selecting
currents in the
reference circuit,
then find the needed
resistances R_1 , R_2 ,
and R_5

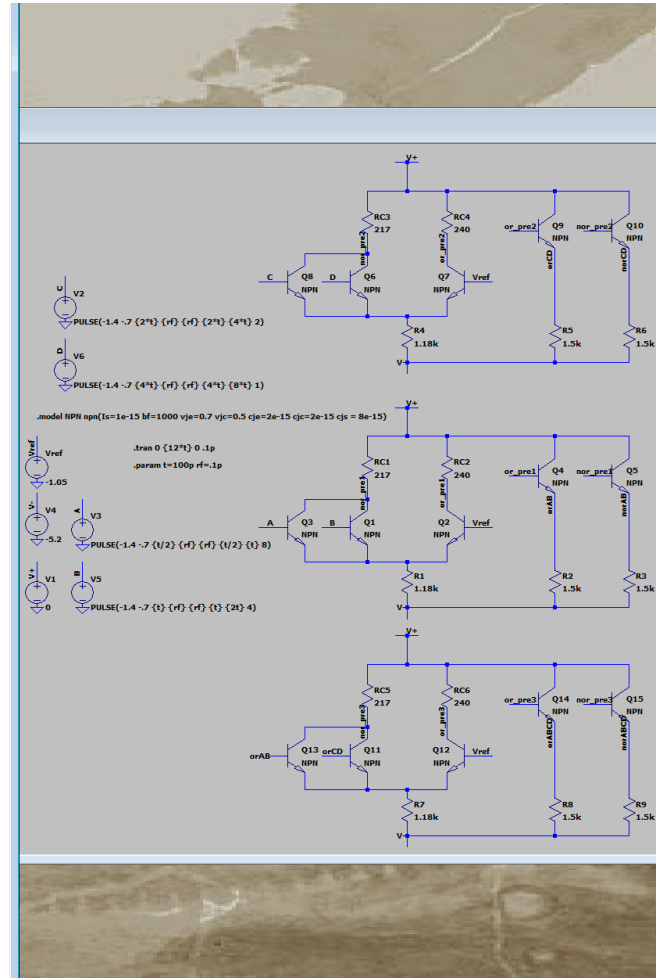
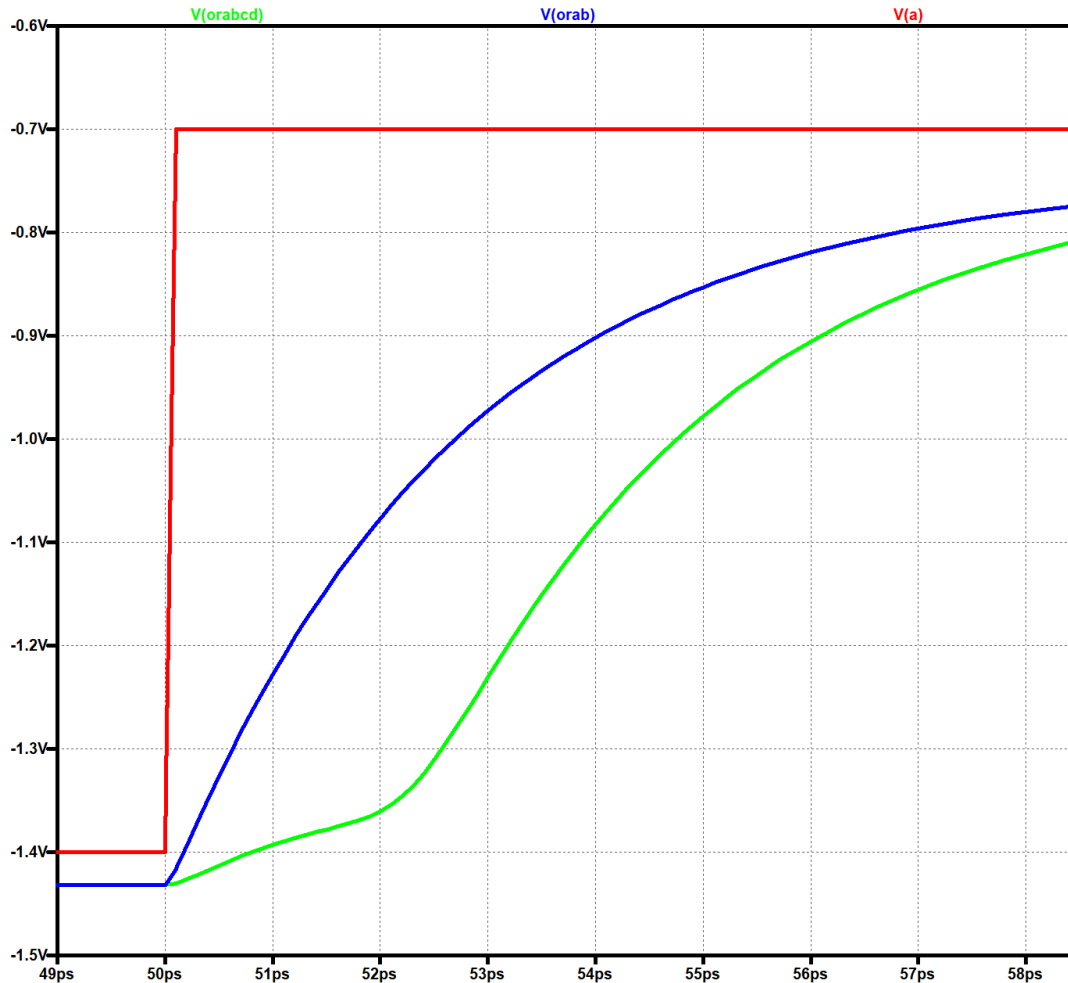
$$P_{DC} = (I_E + I_5 + I_2 + I_3 + I_4) \cdot (V^+ - V^-)$$

Propagation Delay

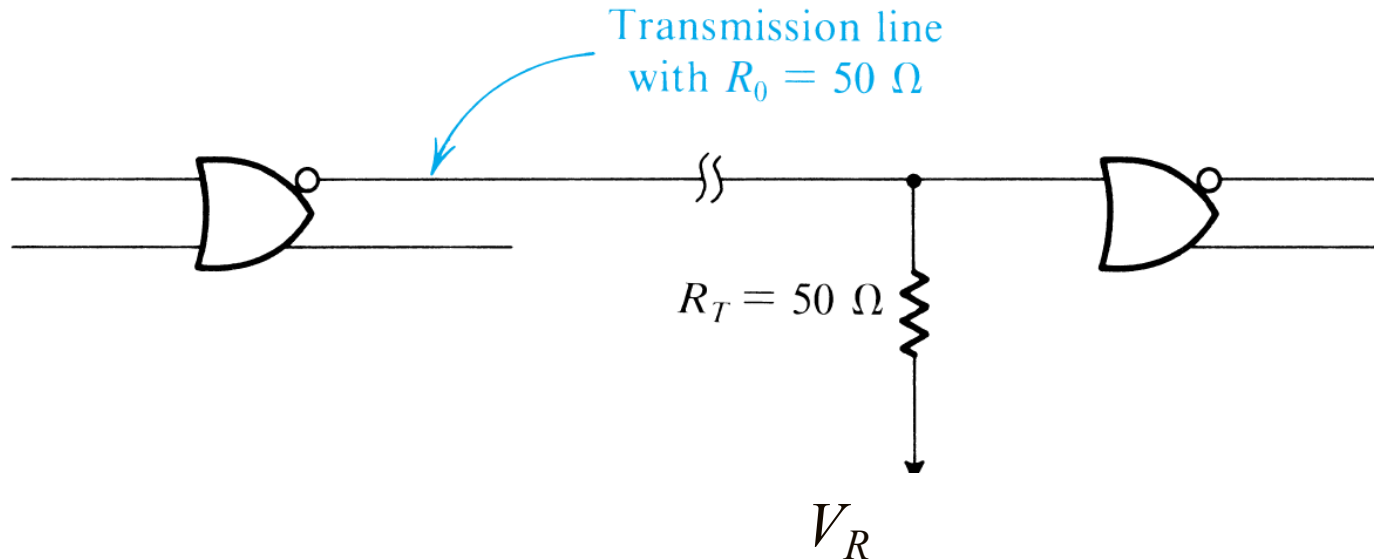
+ The major advantage of ECL circuits is their small propagation delay time (~ 1 ns or less).

- (1) the transistors are not driven into saturation, which eliminates any charge storage effects.
 - (2) the logic swing in the ECL logic gate is small (about 0.7 V), which means that the voltages across the output capacitances do not have to change as much.
 - Currents in the ECL circuit are relatively large, capacitances can charge and discharge quickly.
- High power dissipation
 - Small noise margins.
- +/- ECL circuits are very fast, and they require that special attention be paid to transmission line effects. Improperly designed ECL circuit boards can experience ringing or oscillations → signal line termination required for high-speed circuits.

Propagation Delay NOR4



Termination for High Speed Circuits (10X GHz/10Gbps)

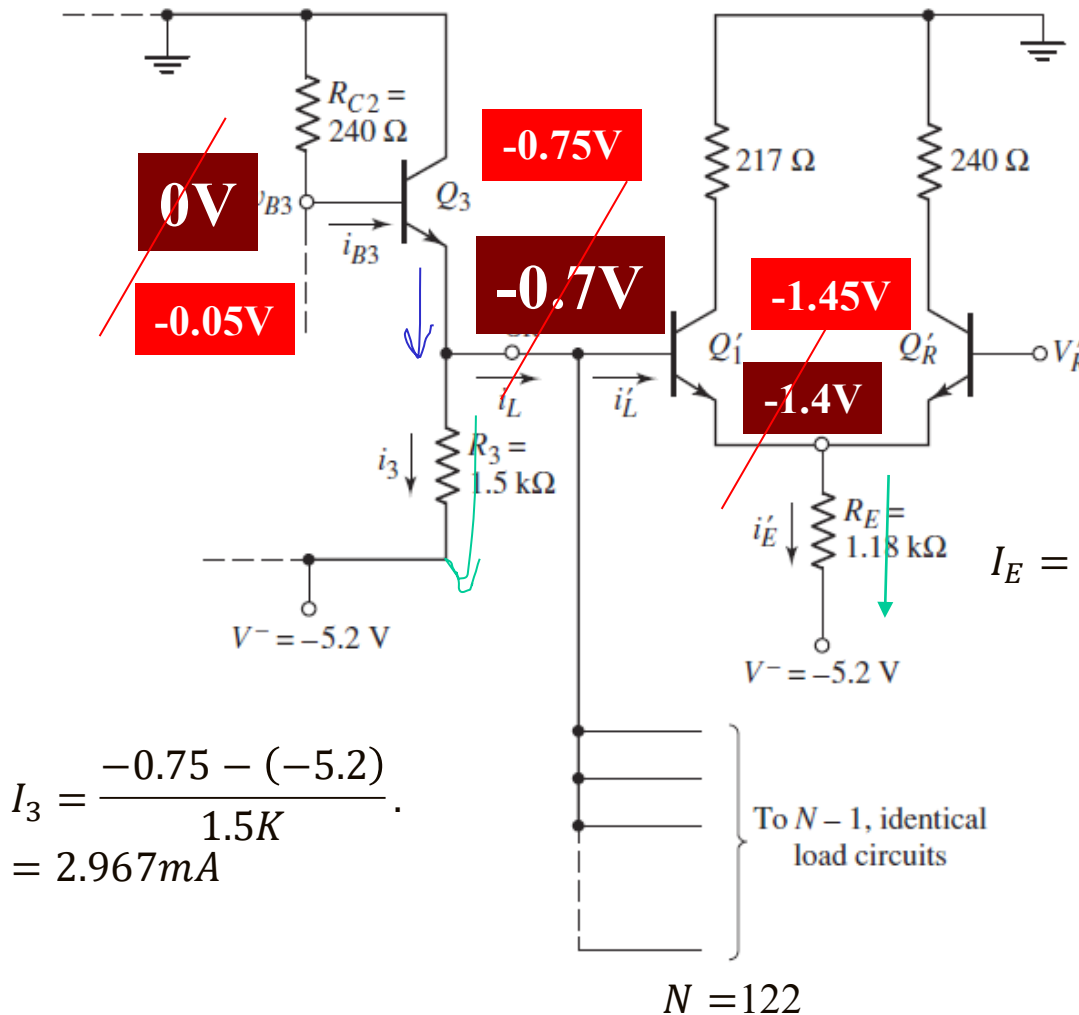


proper way to connect high-speed logic gates such as ECL.
Properly terminating the transmission line connecting the two gates eliminates the "ringing" that would otherwise corrupt the logic signals

Fanout

$$I_{E3} = \frac{0 - (-0.05)}{240} \cdot (1 + 50) = 10.625 \text{mA}$$

Assume $\beta = 50$ as a worst case scenario and that you can go down by 50mV



$$I_3 = \frac{-0.75 - (-5.2)}{1.5K} = 2.967 \text{mA}$$

$$I_E = \frac{-1.45 - (-5.2)}{1.18K} = 3.18 \text{mA}$$

$$I_B = \frac{3.18}{51} = 62.3 \mu\text{A}$$

$$I_{E3} = I_3 + N * 62.3 \mu\text{A}$$

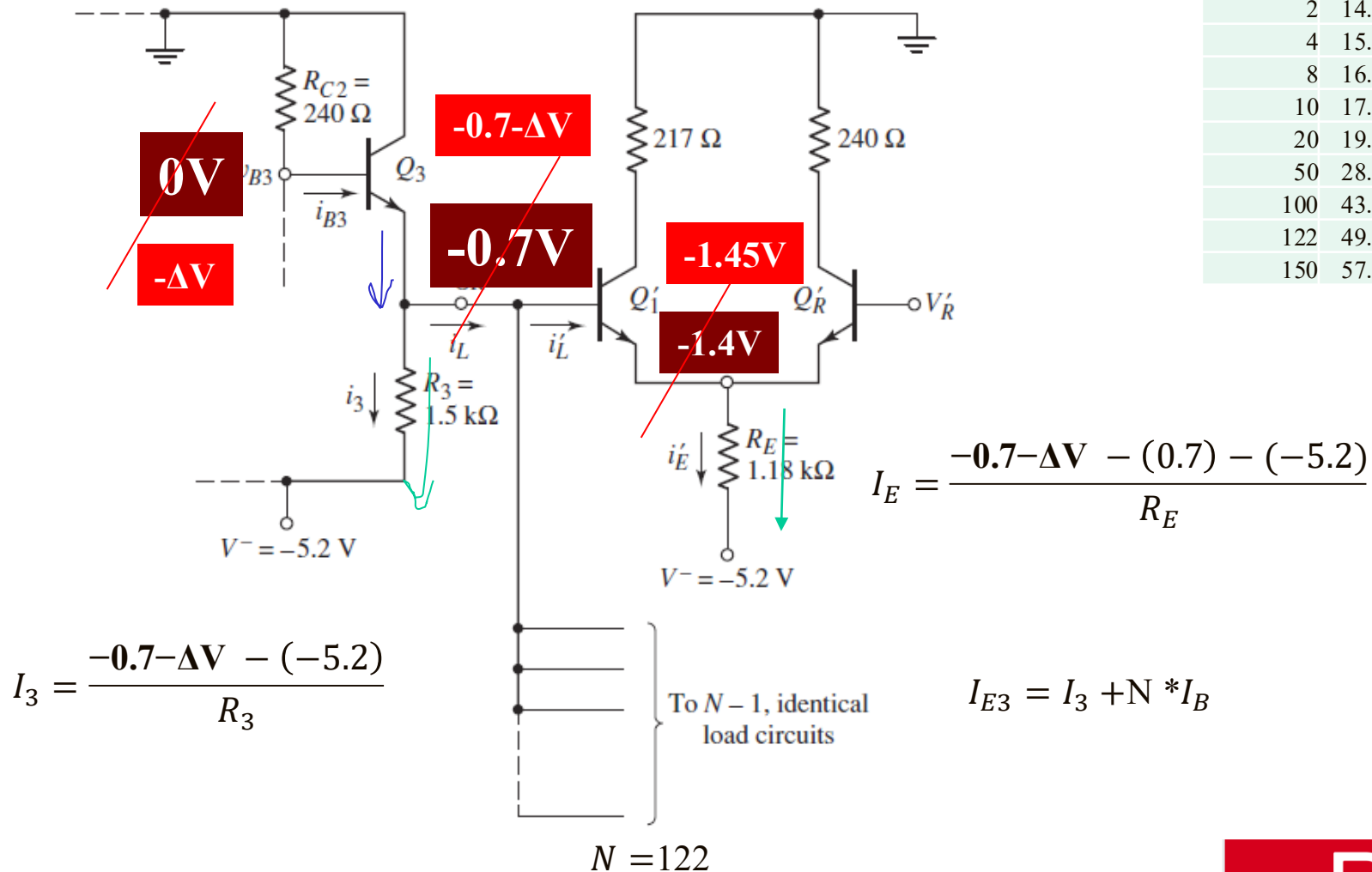
$$7.658 \text{mA} = N * 62.3 \mu\text{A}$$

$$I_{E3} = \frac{\Delta V}{R_{C2}} \cdot (1 + \beta) =$$

Fanout

N ΔV

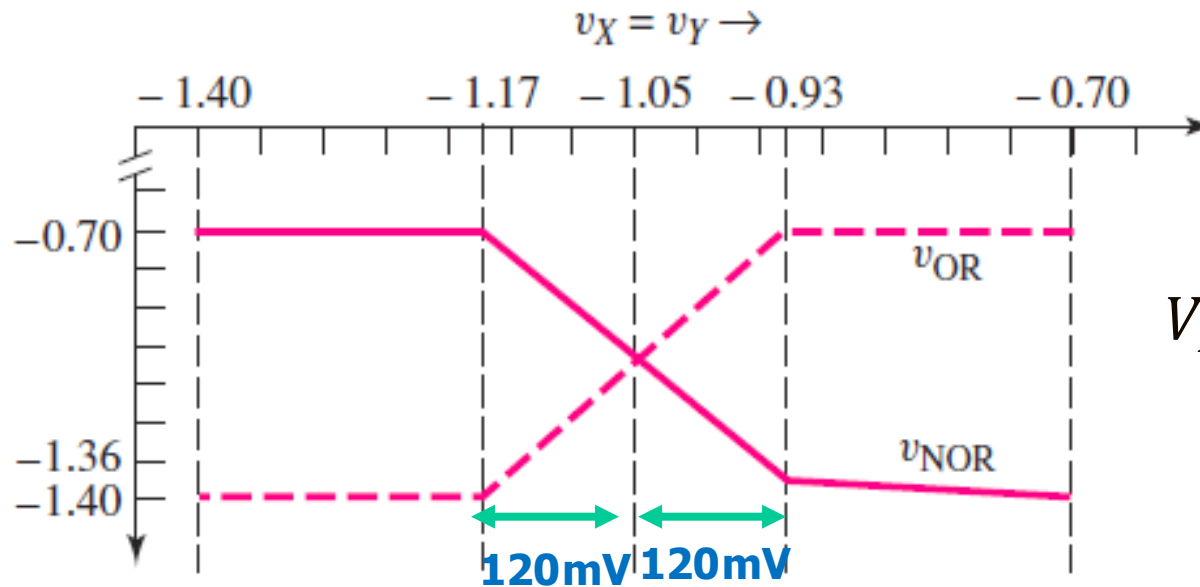
1	14.36859
2	14.66365
4	15.25361
8	16.43299
10	17.02241
20	19.96674
50	28.77227
100	43.35736
122	49.73912
150	57.83007



Comments

- This maximum fanout based on dc conditions is unrealistic.
- In practice, the maximum fanout for ECL circuits is determined by the propagation delay time. Each load circuit increases the load capacitance by approximately x fF (technology dependent).
- A maximum fanout of about 10-15 is usually recommended to keep the propagation delay time within specified limits.

VTC Noise Margin for NOR/OR



$$V_R = \frac{V_{OH} + V_{OL}}{2}$$

for $V^+ = 0$

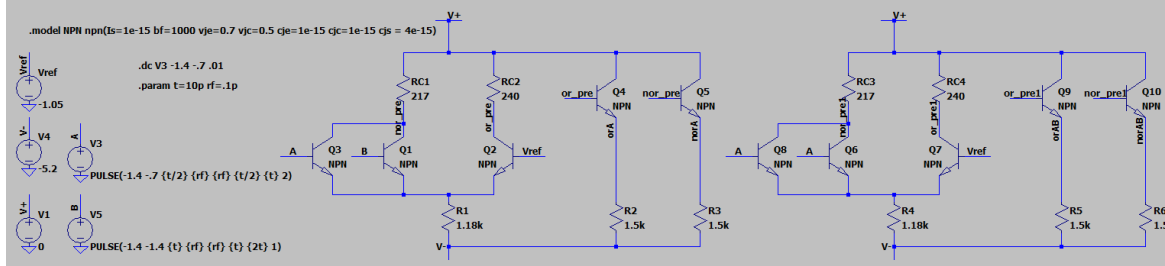
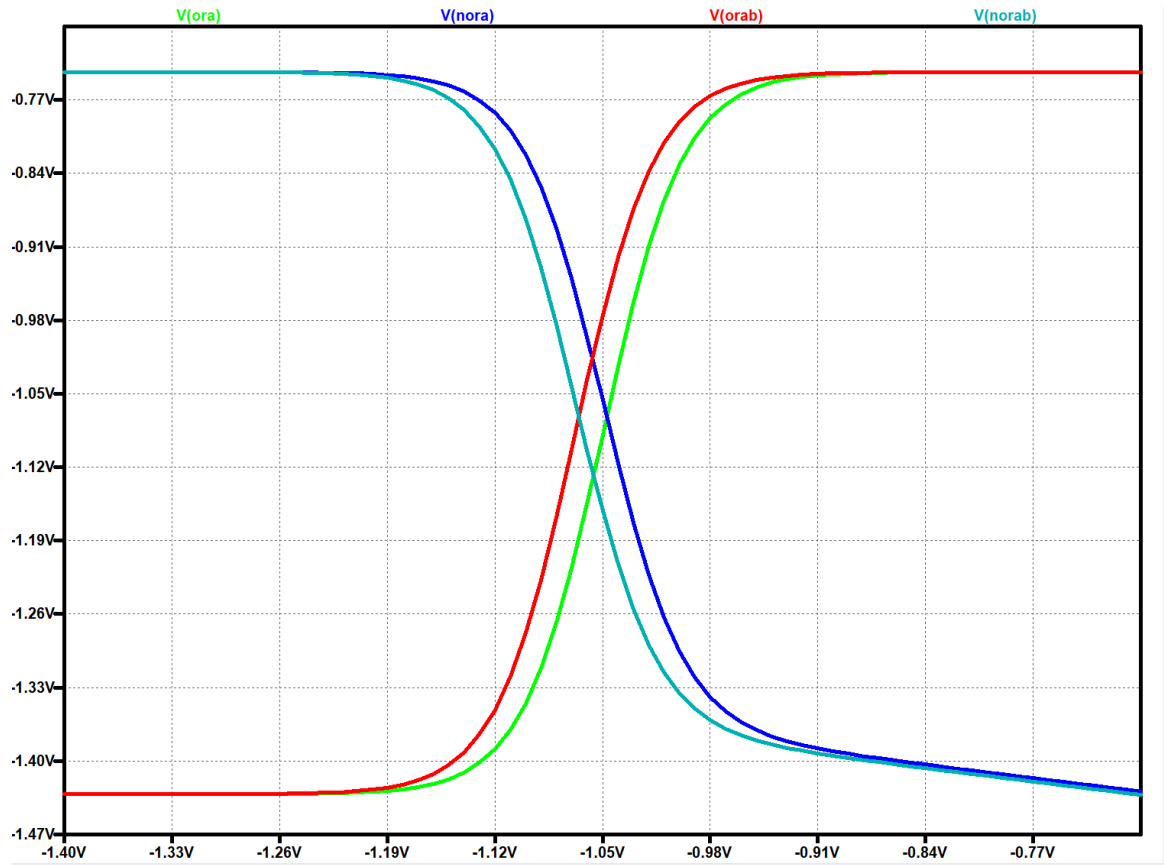
$$V_{OH} = V^+ - 0.7 = -0.7$$

$$V_{OL} = V^+ - 2 \cdot 0.7 = -1.4$$

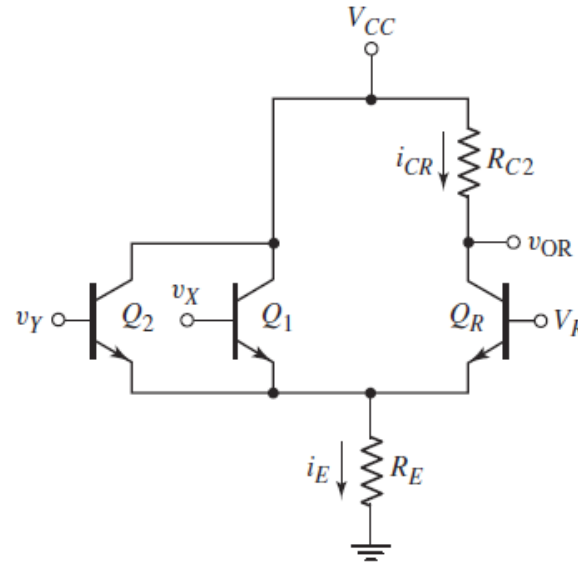
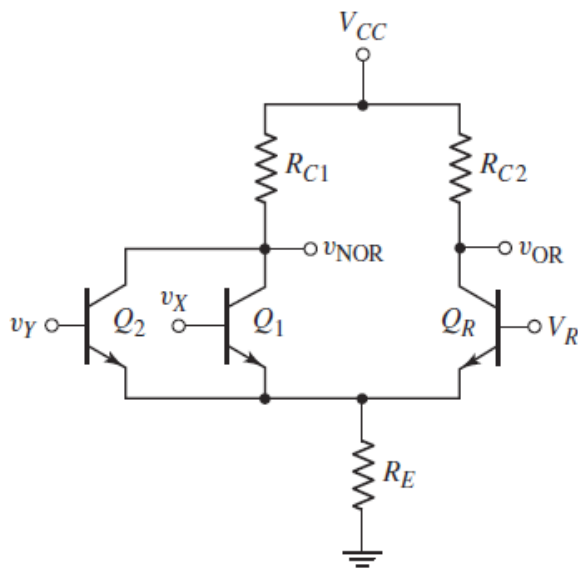
$$V_{IL} = V_R - 5V_T = -1.05 - 0.12 = -1.17\text{V}$$

$$V_{IH} = V_R + 5V_T = -1.05 + 0.12 = -0.93\text{V}$$

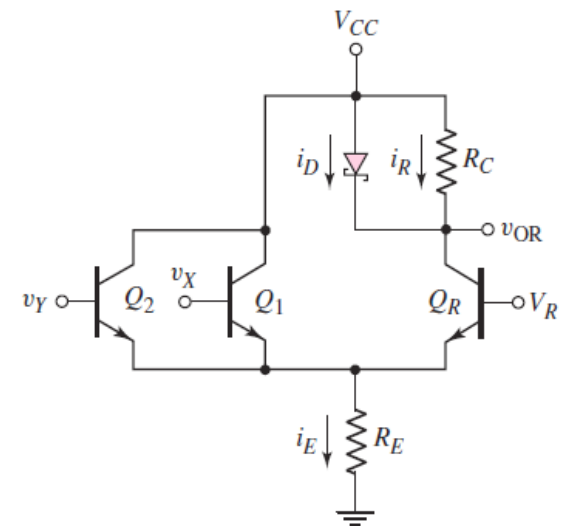
VTC Noise Margin for NOR/OR



Low Power ECL



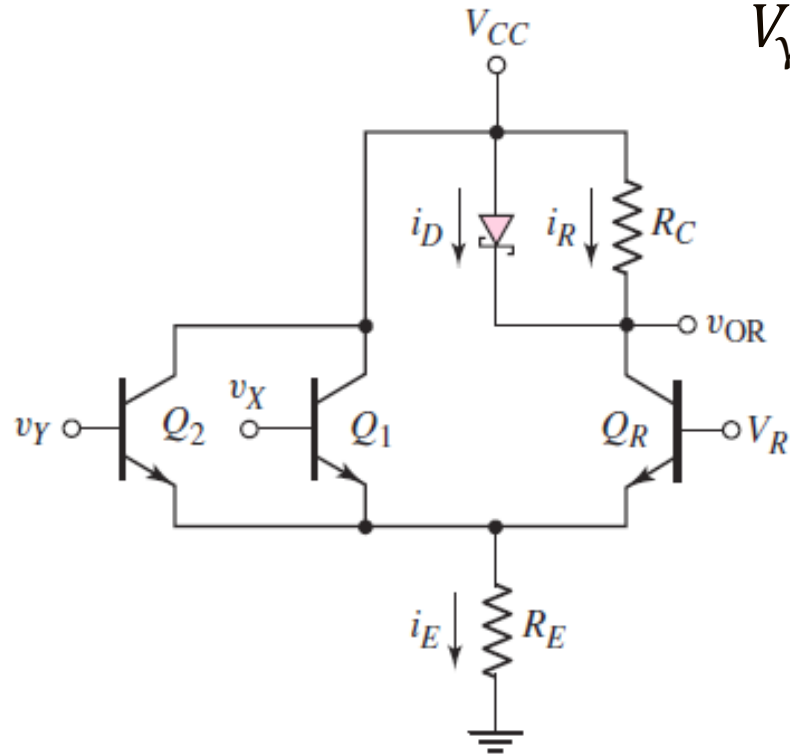
Logic '0' depends on R_{C2}



Logic '0' set by V_{CC} and turn-on voltage of diode

- Make the output voltage states compatible with the input voltages, without the emitter-follower output stages
- If complementary outputs are not required, eliminate resistor R_{C1} (will not affect the power consumption).
- Eliminate the dependence on R_E and R_C

Low Power ECL (OR Gate)



V_Y turn on voltage of the Schottky diode

$$V_x, V_y < V_R \quad Q_1, Q_2 \text{ OFF}$$

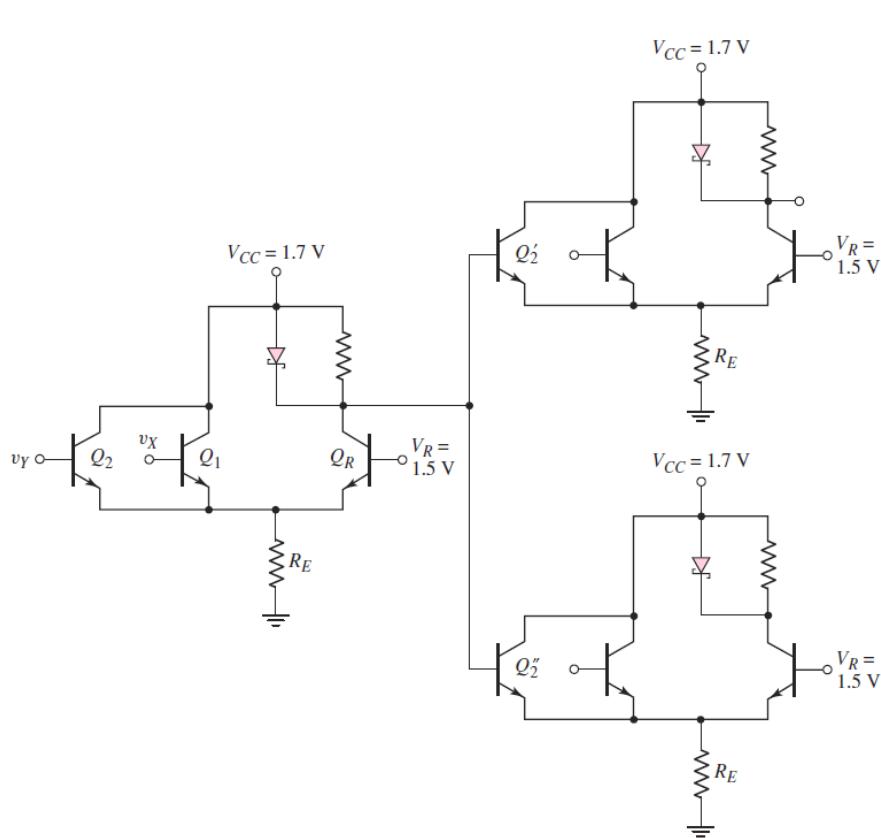
$$V_{OL} = V_{CC} - V_Y$$

$$V_x, V_y > V_R \quad Q_1, Q_2 \text{ ON}$$

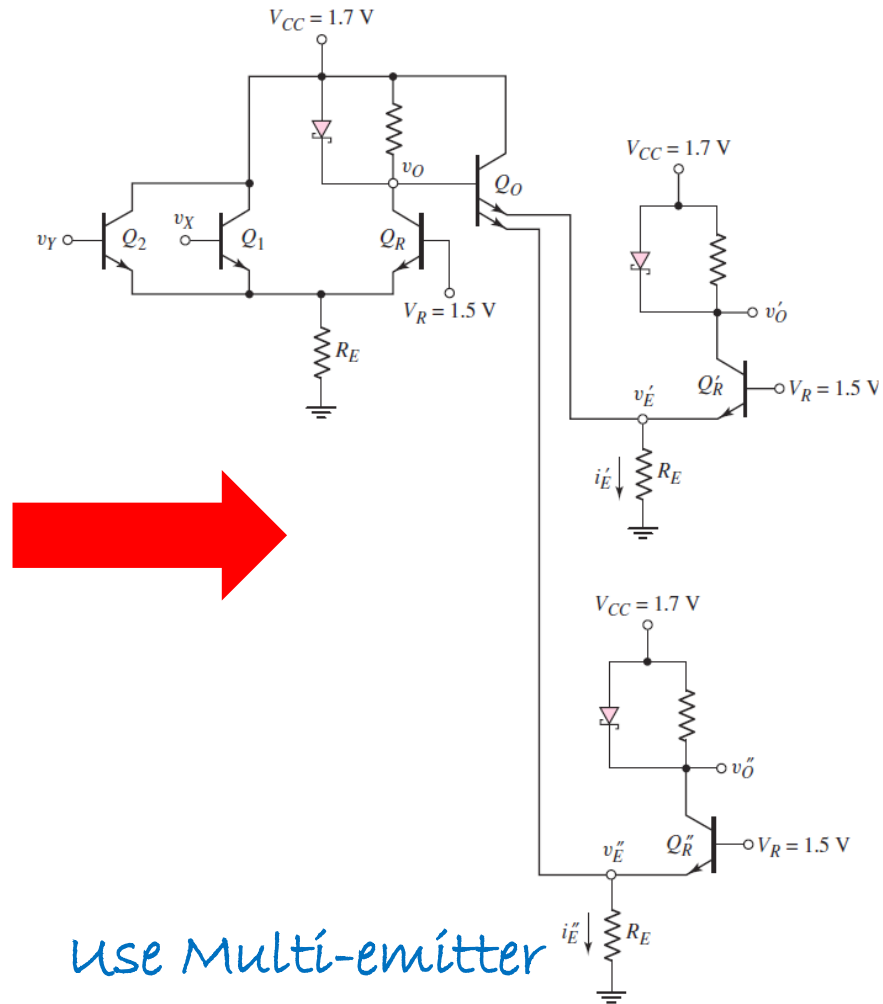
$$V_{OH} = V_{CC}$$

$$V_R = V_{CC} - \frac{V_Y}{2}$$

Modified ECL for IC Implementations



Collectors and bases of Q_2' and Q_2'' are the same

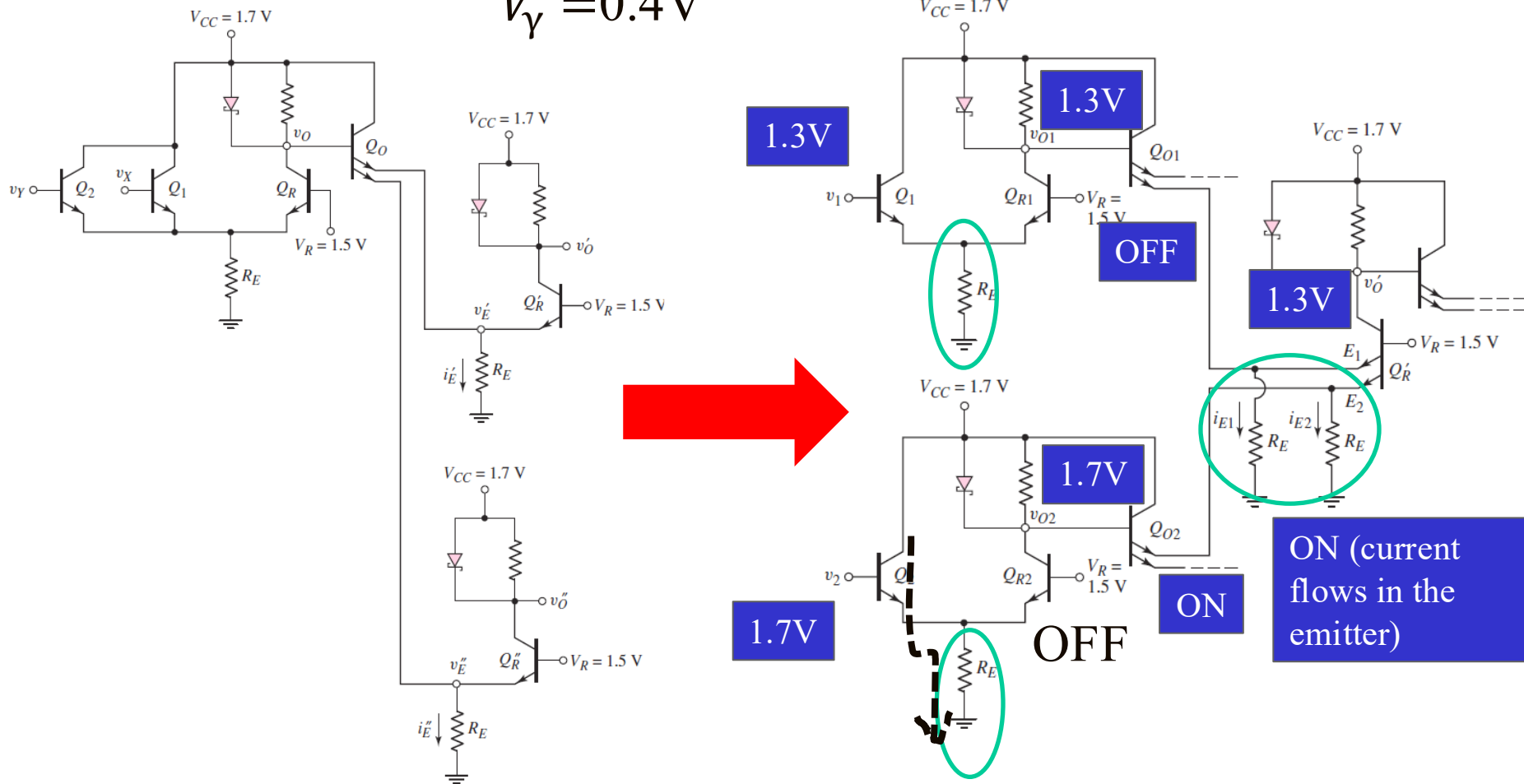


use Multi-emitter transistor

$$V_{\gamma} = 0.4 \text{ V}$$


Modified ECL for IC Implementations

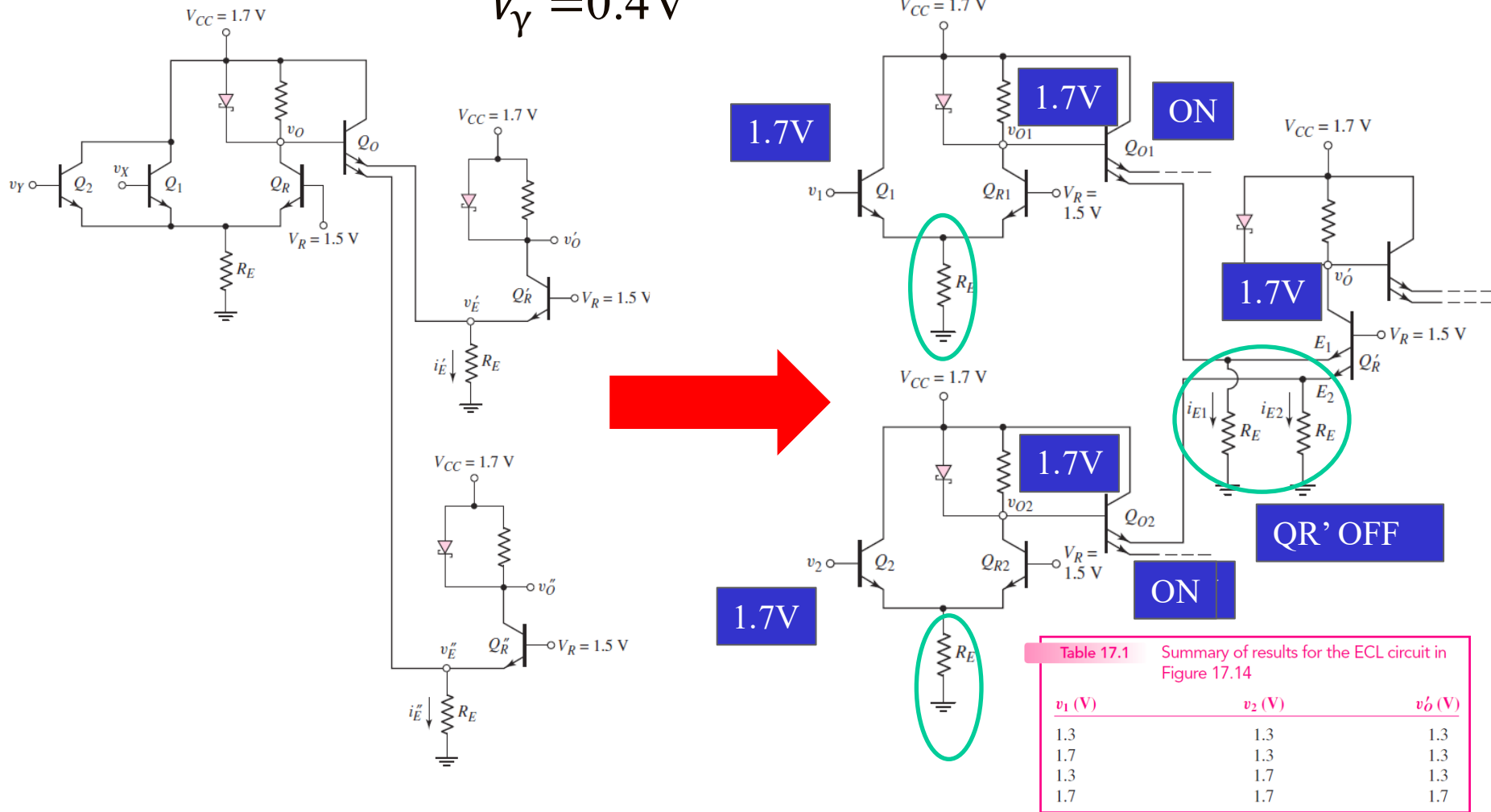
$$V_Y = 0.4V$$



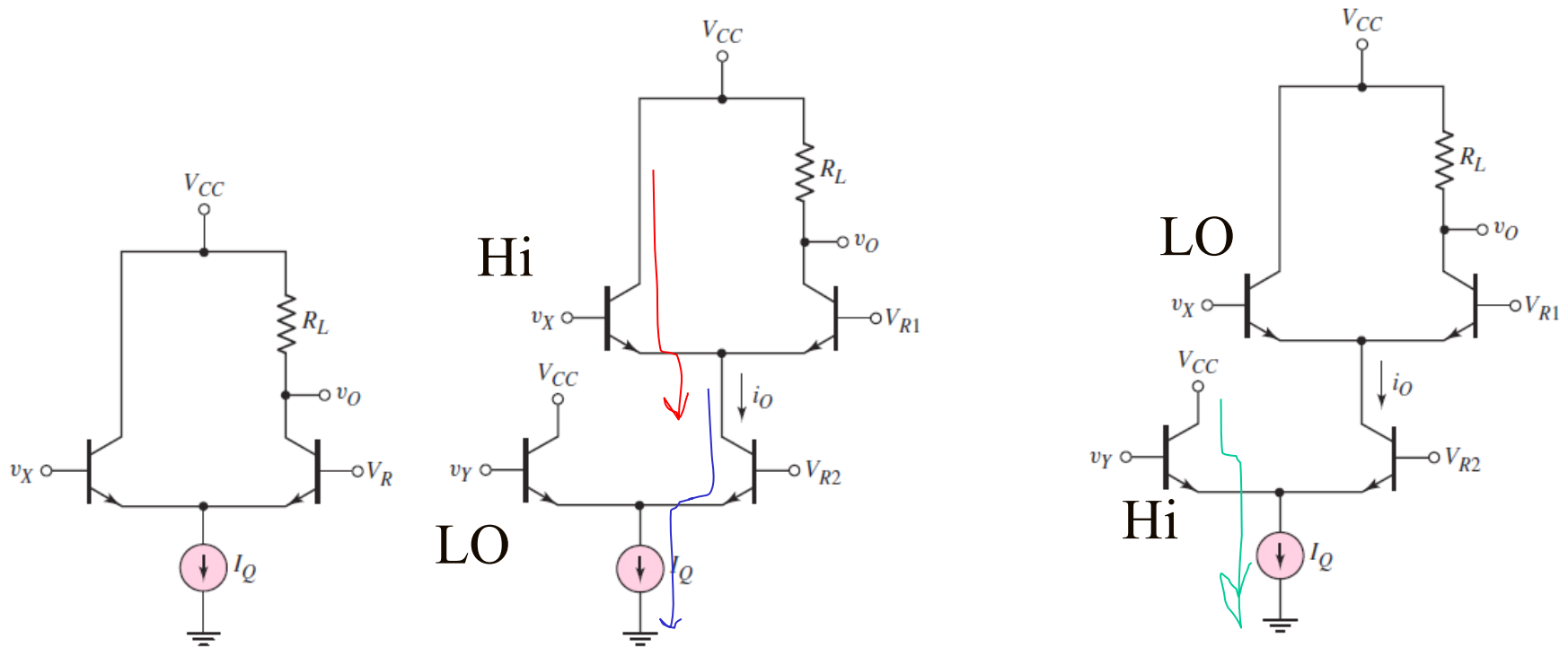
$$V_{OH} = V_{CC} = 1.7V \quad V_{OL} = V_{CC} - V_Y = 1.3V$$

Modified ECL for IC Implementations

$$V_Y = 0.4\text{V}$$

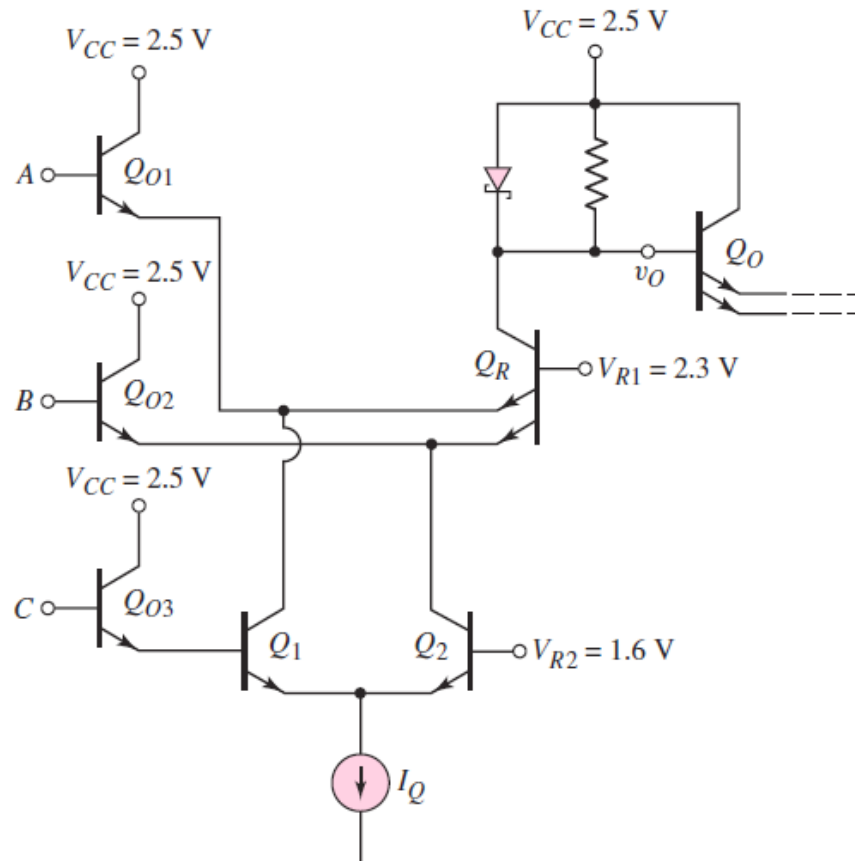


Series Gating-CASCODE



v_X	v_Y	v_O
LO	LO	$V_{CC} - I_Q R_L$
LO	Hi	V_{CC}
Hi	LO	V_{CC}
Hi	Hi	V_{CC}

Series Gating



$(A \text{ AND } C) \text{ OR } (B \text{ AND } \bar{C})$

Propagation Delay time

- ECL is the fastest bipolar logic technology.
- Bipolar technology can produce small, very fast transistors with cutoff frequencies in the range of 500GHz → 700GHz → 1THz.
- Logic gates that use these transistors are so fast that interconnect line delays tend to dominate the propagation delay times.
- Minimizing these interconnect delays involves minimizing the metal lengths and using sufficient current drive capability.
- Speed is derived from low-signal logic swings, non-saturating logic, and the ability to drive a load capacitance.
- Emitter follower? Why?

